

Nernst Equation

"Dependence of electrode Potential and EMF on concentrations and Temperature of an electrode at non standard condition."

The electrode potential of an electrochemical cell depends on the concentration of the electrolyte solution as well as on the temp. Nernst gave the relationship between

electrode potentials and the concentration and temp of an electrolyte solution under study and is k as the Nernst equation.

i.e. In study of the electrochemical cells the electrode potential of the electrode as well as EMF of the cell is determined under standard condition (i.e., electrolyte conc. of each half cell is 1M and temp is 298 K absolute temp)

However, these values will not hold good in case of concentration of electrolyte in two half cells are different from 1M and if temp is also not 298K. Under these conditions the electrode potential of an electrode as well as the EMF of the cell is determined with the help of Nernst equation.

Derivation of Nernst equation

As we know that

$$\Delta G = -nF E_{\text{cell}}$$

$$\Delta G^\circ = -nF E^\circ_{\text{cell}}$$

At standard condition

where conc = 1M

temp = 298K

We also know that

Acc. to thermodynamics

$$\Delta G = \Delta G^\circ + 2.303RT \log Q$$

where,

ΔG = change in Gibbs free energy at

ΔG° = standard condition

$R =$ Universal Gas Constant
 $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$

$T =$ absolute Temp (298K)

$Q =$ Reaction Quotient.

$$Q = \frac{[\text{Product}]}{[\text{Reactant}]}$$

Put the value of $\Delta G + \Delta G^\circ$ in above equation

$$-nFE_{\text{cell}} = -nFE^\circ_{\text{cell}} + 2.303RT \log Q$$

divide both the side by $-nF$

$$\frac{-nFE_{\text{cell}}}{-nF} = \frac{-nFE^\circ_{\text{cell}}}{-nF} + \frac{2.303RT \log Q}{-nF}$$

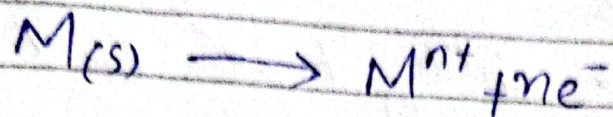
$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{2.303RT \log Q}{nF}$$

$$= E^\circ_{\text{cell}} - \frac{2.303 \times 8.314 \times 298 \log Q}{n \times 96500}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log Q$$

Nernst Equation

Suppose a general oxidation reaction.

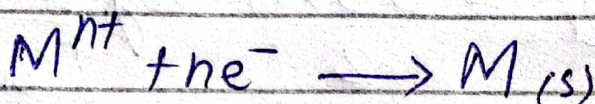


The Nernst equation is written as:

$$E_{\text{ox or cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log \frac{[M^{n+}]}{[M]}$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log [M^{n+}]$$

for reduction reaction

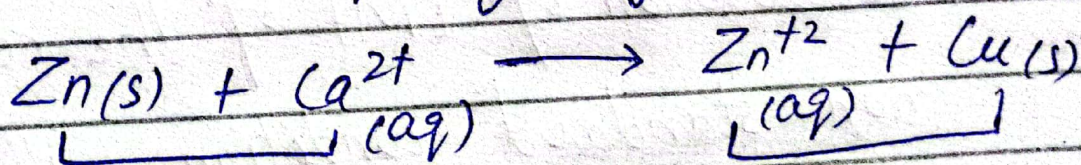


$$E_{\text{rp or cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log \frac{[M]}{[M^{n+}]}$$

Solid phase conc. will always be taken as unity (1)

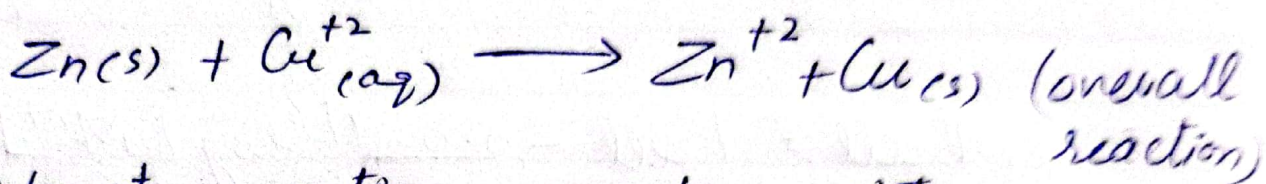
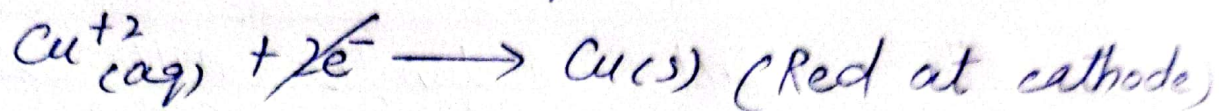
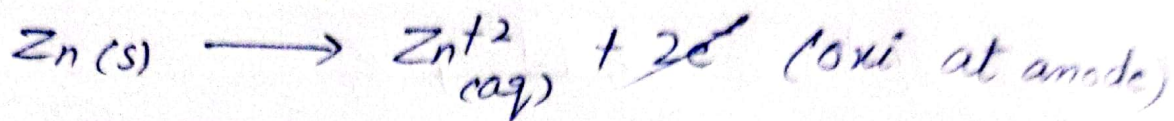
$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log \frac{1}{[M^{n+}]}$$

Write Nernst equ for given reaction



Half cell reaction

At anode:



Nernst equation may be written as:

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Zn}^{+2}][\text{Cu}]}{[\text{Cu}^{+2}][\text{Zn}]}$$

$$[\text{Cu}] = 1$$

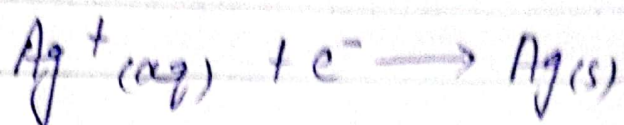
$$[\text{Zn}] = 1$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Zn}^{+2}]}{[\text{Cu}^{+2}]}$$

Application:

Calculation of EMF of the cell at non-standard condition with the help of Nernst equation.

Q: Calculate the reduction potential for the following half cell reaction at 298K.



Given that

$$[\text{Ag}^+] = 0.1\text{M}$$

$$E^\circ_{\text{cell}} = +0.80\text{V}$$

Acc. to Nernst equation.

$$\begin{aligned} E_{\text{cell}} &= E^\circ_{\text{cell}} - \frac{0.0591}{n} \log \frac{1}{[\text{Ag}^+]} \\ &= 0.80 - \frac{0.0591}{n} \log \frac{1}{0.1} \end{aligned}$$

$$\begin{aligned} &\text{as } \log \frac{1}{0.1} = 1 \\ &= 0.80 - \frac{0.0591}{1} \log \frac{1}{0.1} \\ &= 0.80 - 0.0591 \end{aligned}$$

$$E_{\text{cell}} = 0.741\text{V.}$$